

Figure 2

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#ENV

```
rm(list = ls())
setwd("/Users/liqun/Desktop/scMutraceReproducibility/Figure2/")
```

1 Libraries

```
library(ggplot2)
library(ggpubr)
library(ggrepel)
library(dplyr)
##
## Attaching package: 'dplyr'
## The following objects are masked from 'package:stats':
##
##   filter, lag
## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
library(tidyr)
```

```

library(ComplexHeatmap)
## Loading required package: grid
## =====
## ComplexHeatmap version 2.24.1
## Bioconductor page: http://bioconductor.org/packages/ComplexHeatmap/
## Github page: https://github.com/jokergoo/ComplexHeatmap
## Documentation: http://jokergoo.github.io/ComplexHeatmap-reference
##
## If you use it in published research, please cite either one:
## - Gu, Z. Complex Heatmap Visualization. iMeta 2022.
## - Gu, Z. Complex heatmaps reveal patterns and correlations in multidimensional
##   genomic data. Bioinformatics 2016.
##
##
## The new InteractiveComplexHeatmap package can directly export static
## complex heatmaps into an interactive Shiny app with zero effort. Have a try!
##
## This message can be suppressed by:
##   suppressPackageStartupMessages(library(ComplexHeatmap))
## =====
library(circlize)
## =====
## circlize version 0.4.16
## CRAN page: https://cran.r-project.org/package=circlize
## Github page: https://github.com/jokergoo/circlize
## Documentation: https://jokergoo.github.io/circlize\_book/book/
##
## If you use it in published research, please cite:
## Gu, Z. circlize implements and enhances circular visualization
##   in R. Bioinformatics 2014.
##
## This message can be suppressed by:
##   suppressPackageStartupMessages(library(circlize))
## =====

```

2 def function

```
# def function
```

3 Load Data

```

# XVSeq
# Ultraviolet radiation shapes dendritic cell leukaemia transformation in the skin
# https://www.nature.com/articles/s41586-023-06156-8
# Mutation_sample_XVSeq <- read.csv("../Data/XVSeq/Mutation_sample.csv", row.names = 1)
# Mutation_sample_XVSeq[is.na(Mutation_sample_XVSeq)] <- 0
# Mutation_sample_XVSeq$Gene <- rownames(Mutation_sample_XVSeq)

```

```

# GoTSeq
# Deconvoluting clonal and cellular architecture in IDH-mutant acute myeloid leukemia
# https://www.cell.com/cell-stem-cell/fulltext/S1934-5909\(25\)00179-1
# Mutation_sample_GoTSeq <- read.csv("../Data/GoT/Mutation_sample.csv", row.names = 1)
# Mutation_sample_GoTSeq[is.na(Mutation_sample_GoTSeq)] <- 0
# Mutation_sample_GoTSeq$Gene <- rownames(Mutation_sample_GoTSeq)

# SMARTSeq
# Therapy-Induced Evolution of Human Lung Cancer Revealed by Single-Cell RNA Sequencing
# https://www.sciencedirect.com/science/article/pii/S0092867420308825
# Mutation_sample_SMARTSeq <- read.csv("../Data/SMARTSeq/Mutation_sample.csv", row.names = 1)
# Mutation_sample_SMARTSeq[is.na(Mutation_sample_SMARTSeq)] <- 0
# Mutation_sample_SMARTSeq$Gene <- rownames(Mutation_sample_SMARTSeq)

# Merge
# Mutation_sample_XVSeq <- Mutation_sample_XVSeq |> dplyr::rename_with(~ paste0("XVSeq_", .), -Gene)
# Mutation_sample_GoTSeq <- Mutation_sample_GoTSeq |> dplyr::rename_with(~ paste0("GoTSeq_", .), -Gene)
# Mutation_sample_SMARTSeq <- Mutation_sample_SMARTSeq |> dplyr::rename_with(~ paste0("SMARTSeq_", .), -Gene)

# Merged_mutation <- Mutation_sample_XVSeq %>%
#   dplyr::full_join(Mutation_sample_GoTSeq, by = "Gene") %>%
#   dplyr::full_join(Mutation_sample_SMARTSeq, by = "Gene")

# Merged_mutation[is.na(Merged_mutation)] <- 0

# save merge data
# write.csv(Merged_mutation, "../Data/0_Mutation_sample_merge.csv")

# load Mutation sample matrix
Merged_mutation <- read.csv("../Data/0_Mutation_sample_merge.csv")

# load Mutant cell fraction
Mutation_fraction <- read.csv("../Data/0_Benchmarking_MutCellFrac.csv", header = T)

rownames(Merged_mutation) <- Merged_mutation$Gene
Merged_mutation_Ann <- Merged_mutation[, c("Gene", "Tag")]
Merged_mutation$Gene <- NULL
Merged_mutation$Tag <- NULL

# heatmap
mat <- as.matrix(Merged_mutation)
bin_mat <- ifelse(mat > 0, 1, 0)

col_fun2 <- c("0" = "#202020",
              "1" = "#ca0020")

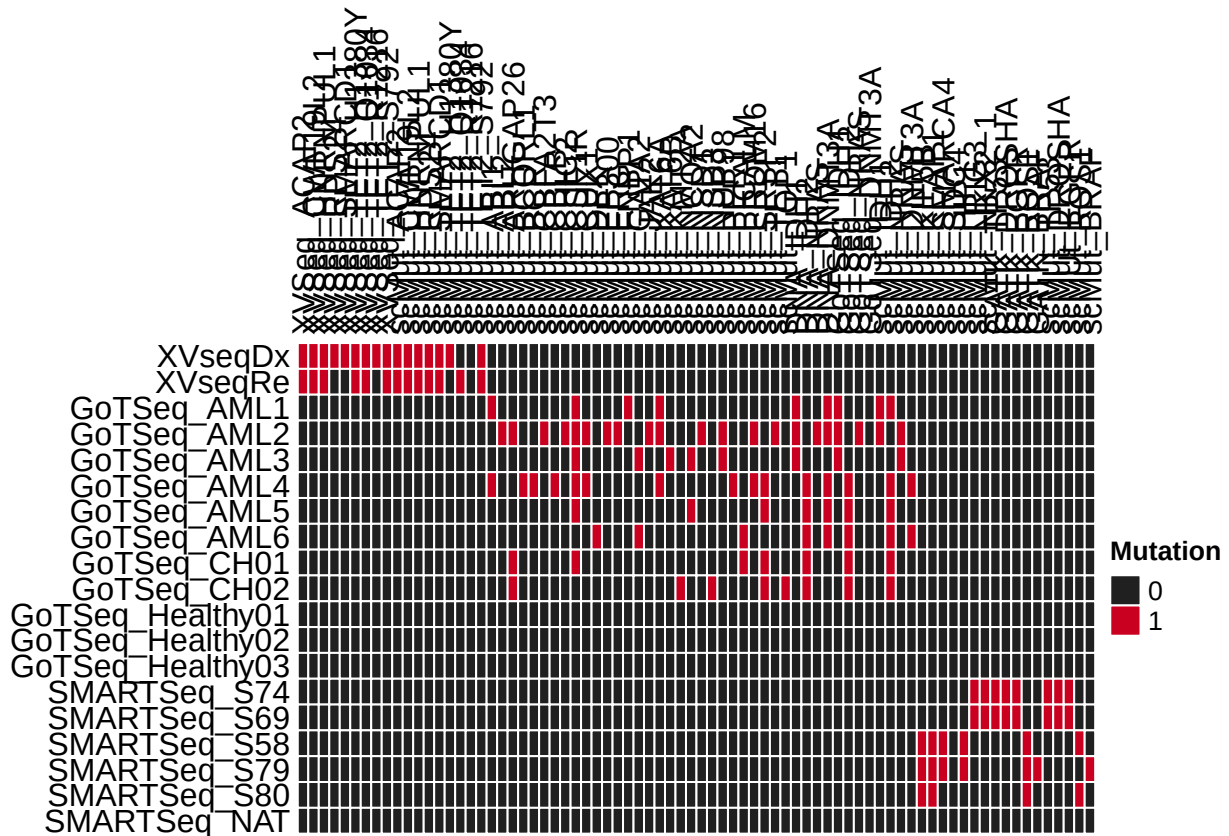
Heatmap(
  t(bin_mat),
  name = "Mutation",
  col = col_fun2,
  cluster_rows = FALSE,
  cluster_columns = FALSE,
  row_names_side = "left",

```

```

column_names_side = "top",
rect_gp = gpar(col = "white", lwd = 1)
)

```



```

p_correlation_scMut_OtherMethods <- ggplot(Mutation_fraction, aes(x = scMutrace, y = OtherMethod)) +
  geom_point(size = 1.5, color = "#202020") +
  geom_smooth(method = "lm", se = TRUE, color = "#ca0020", linetype = "dashed", lwd = 1.2) +
  theme_classic(base_size = 16) +
  xlab("scMutrace") +
  ylab("OtherMethod") +
  xlim(0, 1) + ylim(0, 0.6) +
  annotate("text",
    x = max(Mutation_fraction$scMutrace) * 0.7,
    y = max(Mutation_fraction$OtherMethod) * 0.15,
    label = paste0("r = ", round(cor(Mutation_fraction$scMutrace, Mutation_fraction$OtherMethod), 2)),
    size = 5)

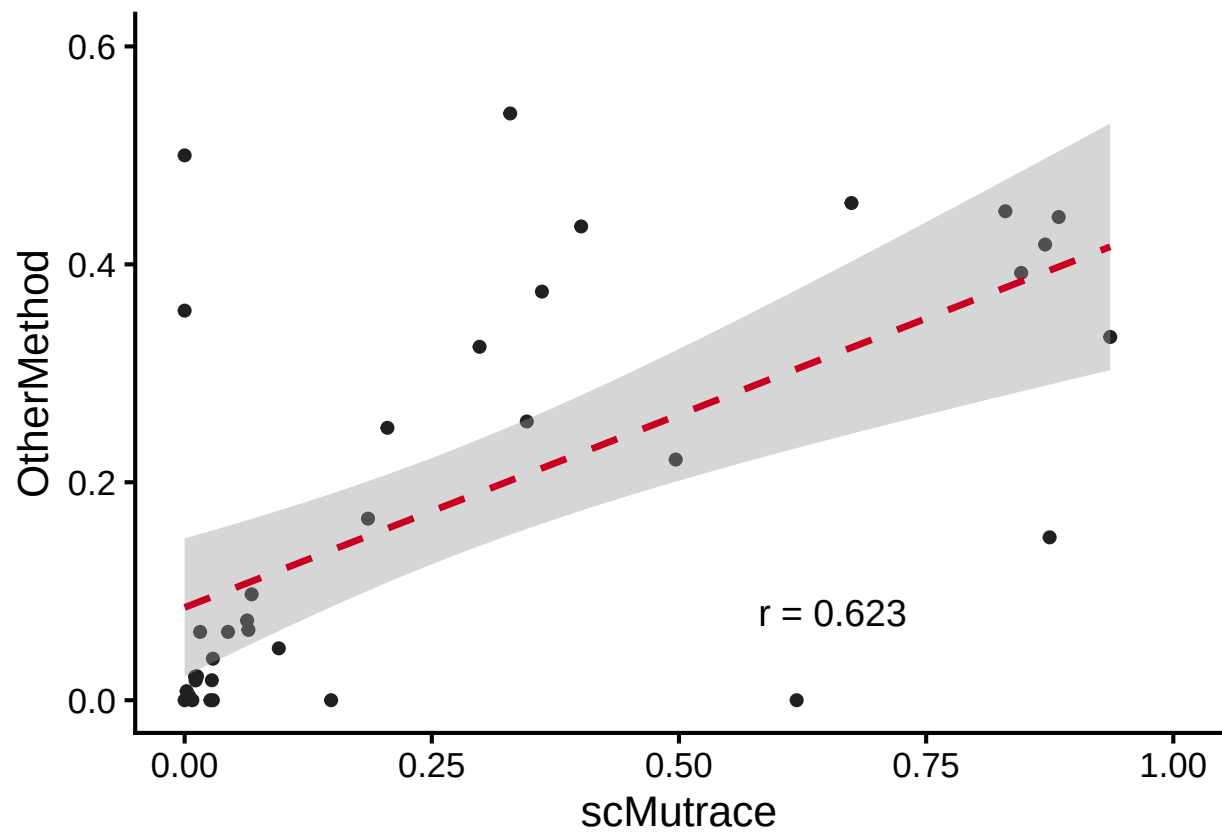
cor.test(Mutation_fraction$scMutrace, Mutation_fraction$OtherMethod, method = "pearson")
##
## Pearson's product-moment correlation
##
## data: Mutation_fraction$scMutrace and Mutation_fraction$OtherMethod
## t = 4.7132, df = 35, p-value = 3.803e-05
## alternative hypothesis: true correlation is not equal to 0

```

```
## 95 percent confidence interval:
## 0.3747529 0.7880265
## sample estimates:
##      cor
## 0.6231096

pdf("../Figures/p_Mutation_Sample_heatmap.pdf", width = 14.54, height = 4.88)
Heatmap(
  t(bin_mat),
  name = "Mutation",
  col = col_fun2,
  cluster_rows = FALSE,
  cluster_columns = FALSE,
  row_names_side = "left",
  column_names_side = "top",
  rect_gp = gpar(col = "white", lwd = 1)
)
dev.off()
## pdf
## 2

p_correlation_scMut_OtherMethods
## `geom_smooth()` using formula = 'y ~ x'
```



```

ggsave("../Figures/p_correlation_scMut_OtherMethods.pdf", width = 3.08, height = 3.08)
## `geom_smooth()` using formula = 'y ~ x'
# clean mat
rm(mat, bin_mat)

```

4 benchmarking

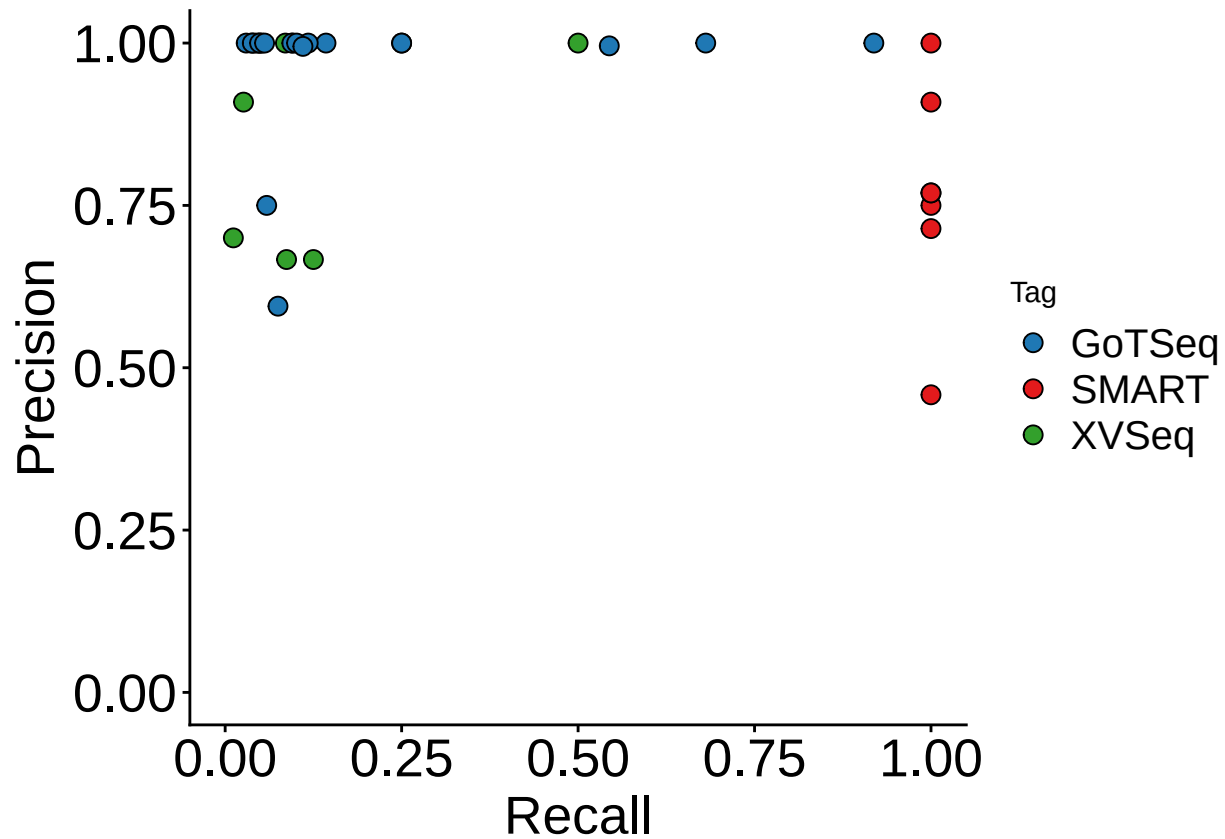
```

benchmarking_data <- read.csv("../Data/0_Benchmarking_Data.csv", header = T)

p_benchmarking_res <- ggplot(data = benchmarking_data) +
  geom_point(mapping = aes(x = Recall, y = Precision, fill = Tag),
    color = "black", shape = 21, size = 3, show.legend = TRUE) +
  scale_fill_manual(values = c("#1f78b4", "#e31a1c", "#33a02c")) +
  xlab("Recall") + ylab("Precision") + xlim(0, 1) + ylim(0, 1) +
  theme_classic() +
  theme(
    axis.title = element_text(size = 20),
    axis.text.y = element_text(size = 20),
    axis.text.x = element_text(size = 20),
    legend.text = element_text(size = 15)
  )

p_benchmarking_res

```



```
ggsave("../Figures/p_benchmarking_res.pdf", p_benchmarking_res, width = 4.9, height = 3.08)
```

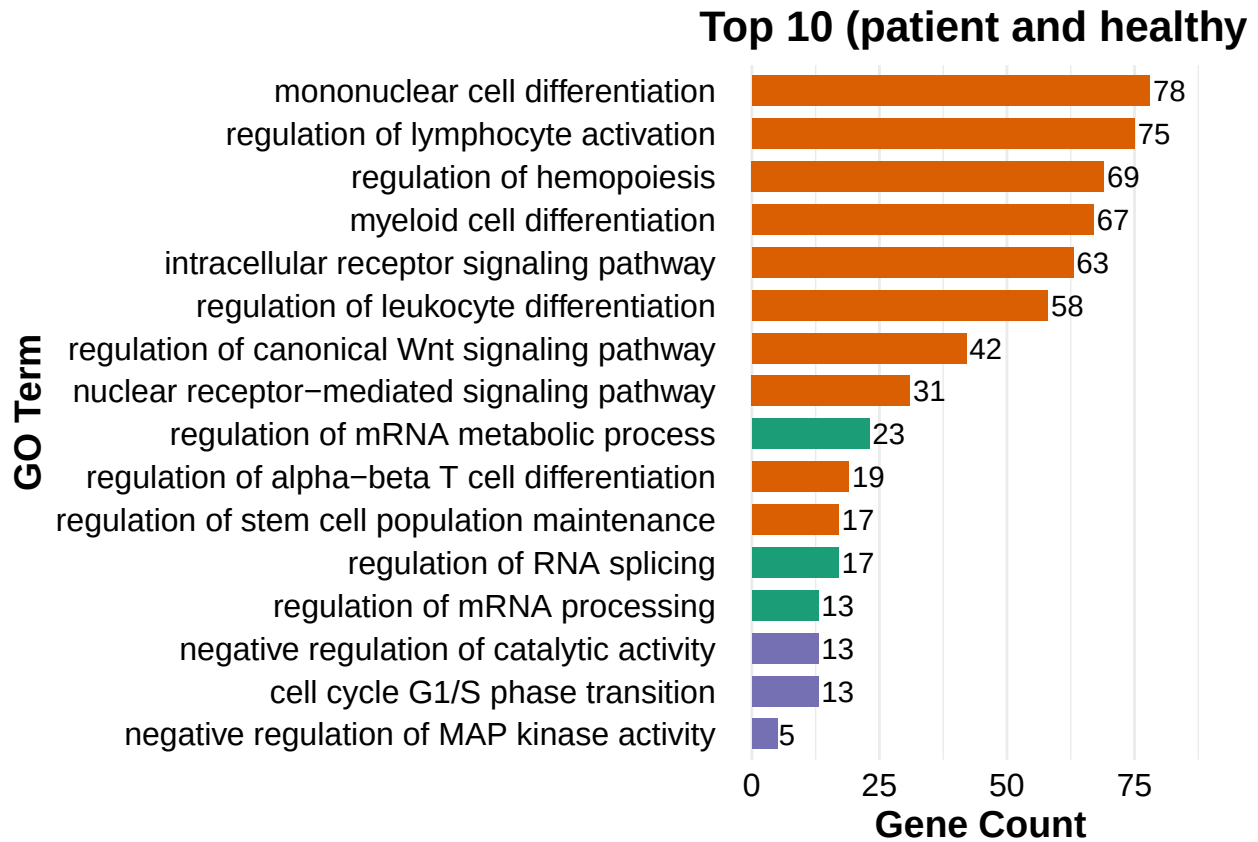
5 GO

```
GO_mutation <- read.csv("../Data/1_GO.csv", header = T)

p_GOs_mutation <- ggplot(GO_mutation, aes(x = reorder(Description, Count), y = Count, fill = Type)) +
  geom_bar(stat = "identity", width = 0.7) +
  coord_flip() +
  geom_text(aes(label = Count), hjust = -0.1, size = 4, color = "black") +
  labs(
    x = "GO Term",
    y = "Gene Count",
    title = "Top 10 (patient and healthy)"
  ) + theme_minimal() +
  theme(
    axis.text = element_text(size = 12, color = "black"),
    axis.title = element_text(size = 14, face = "bold"),
    plot.title = element_text(size = 16, face = "bold", hjust = 0.5),
    legend.position = "none",
    panel.grid.major.y = element_blank()
  ) + ylim(0, max(GO_mutation$Count)*1.15) +
  scale_fill_manual(values = c(
```

```
"AML" = "#d95f02",
"CH" = "#1b9e77",
"Healthy" = "#7570b3"))
```

p_GOs_mutation



```
ggsave("../Figures/p_GOs_mutation.pdf", p_GOs_mutation, width = 5.20, height = 4.21)
```

6 SBS mutation signatures

```
SBS_data <- read.csv("../Data/XVSeq/Signature_contributions_PDC.csv", header = T)
df_long <- SBS_data %>%
  tidyr::pivot_longer(cols = c(DX, RE),
    names_to = "Stage",
    values_to = "Contribution")

signature_colors <- c(
  "SBS1" = "#a6cee3",
  "SBS5" = "#1f78b4",
  "SBS7a" = "#e31a1c",
  "SBS7b" = "#fb9a99",
  "SBS37" = "#737373",
```



```

    "SBS44" = "#f0f0f0"
  )

p_SBS_XVSeq <- ggplot(df_long, aes(x = Stage, y = Contribution, fill = Signature)) +
  geom_bar(stat = "identity", position = "stack") +
  scale_fill_manual(values = signature_colors) +
  theme_bw() +
  labs(
    title = "Mutational Signature Contributions (DX vs RE)",
    x = "Stage",
    y = "Contribution"
  )

ggsave("../Figures/p_SBS_XVSeq.pdf", width = 3.34, height = 4.22)

```

7 sessionInfo

```

sessionInfo()
## R version 4.5.1 (2025-06-13)
## Platform: aarch64-apple-darwin20
## Running under: macOS Tahoe 26.2
##
## Matrix products: default
## BLAS:   /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRblas.0.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib; LAPACK v
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: Europe/Stockholm
## tzcode source: internal
##
## attached base packages:
## [1] grid      stats      graphics  grDevices  utils      datasets  methods
## [8] base
##
## other attached packages:
## [1] circlize_0.4.16      ComplexHeatmap_2.24.1 tidyr_1.3.1
## [4] dplyr_1.1.4          ggrepel_0.9.6         ggpubr_0.6.1
## [7] ggplot2_4.0.0
##
## loaded via a namespace (and not attached):
## [1] gtable_0.3.6      shape_1.4.6.1      rjson_0.2.23
## [4] xfun_0.53         GlobalOptions_0.1.2 rstatix_0.7.2
## [7] lattice_0.22-7    Cairo_1.6-5         vctrs_0.6.5
## [10] tools_4.5.1       generics_0.1.4      stats4_4.5.1
## [13] parallel_4.5.1    tibble_3.3.0        cluster_2.1.8.1
## [16] pkgconfig_2.0.3    Matrix_1.7-4        RColorBrewer_1.1-3
## [19] S7_0.2.0           S4Vectors_0.46.0    lifecycle_1.0.4
## [22] compiler_4.5.1     farver_2.1.2        textshaping_1.0.3
## [25] codetools_0.2-20   carData_3.0-5       clue_0.3-66

```

```
## [28] htmltools_0.5.8.1    yaml_2.3.10          Formula_1.2-5
## [31] pillar_1.11.1        car_3.1-3            crayon_1.5.3
## [34] iterators_1.0.14     abind_1.4-8          foreach_1.5.2
## [37] nlme_3.1-168         tidyselect_1.2.1     digest_0.6.37
## [40] purrr_1.1.0          labeling_0.4.3        splines_4.5.1
## [43] fastmap_1.2.0        colorspace_2.1-2     cli_3.6.5
## [46] magrittr_2.0.4       dichromat_2.0-0.1    broom_1.0.10
## [49] withr_3.0.2          scales_1.4.0         backports_1.5.0
## [52] rmarkdown_2.29       matrixStats_1.5.0    ggsignif_0.6.4
## [55] ragg_1.5.0           png_0.1-8            GetoptLong_1.0.5
## [58] evaluate_1.0.5       knitr_1.50           IRanges_2.42.0
## [61] doParallel_1.0.17    mgcv_1.9-3           rlang_1.1.6
## [64] Rcpp_1.1.0           glue_1.8.0           BiocGenerics_0.54.0
## [67] rstudioapi_0.17.1    R6_2.6.1             systemfonts_1.2.3
```